OCI secondary DNS upstream migration

Customer implementation guide

Bulk upstream updates with per-zone TSIG

Reusable Python workflow for customer DNS operations

15 October 2026

Customer migration scope

Controlled batch updates across selected secondary zones

Selection	A reviewed OCID file defines the exact zones in each batch.
Migration	Add the target during overlap, then remove the source separately.
Validation	Fresh records and transfer evidence support each change gate.
Recovery	Original settings and per-zone results support recovery.

DNS responsibilities

Record ownership and secondary service roles

Component	Responsibility
Source upstream	Maintains the current zone data and SOA serials
Target upstream	Serves matching records with each zone’s TSIG key
OCI secondary DNS	Pulls zone contents and answers authoritative queries
Operator	Runs scripts and reviews change and transfer evidence

Migration stages

A separate verification gate follows each configuration change

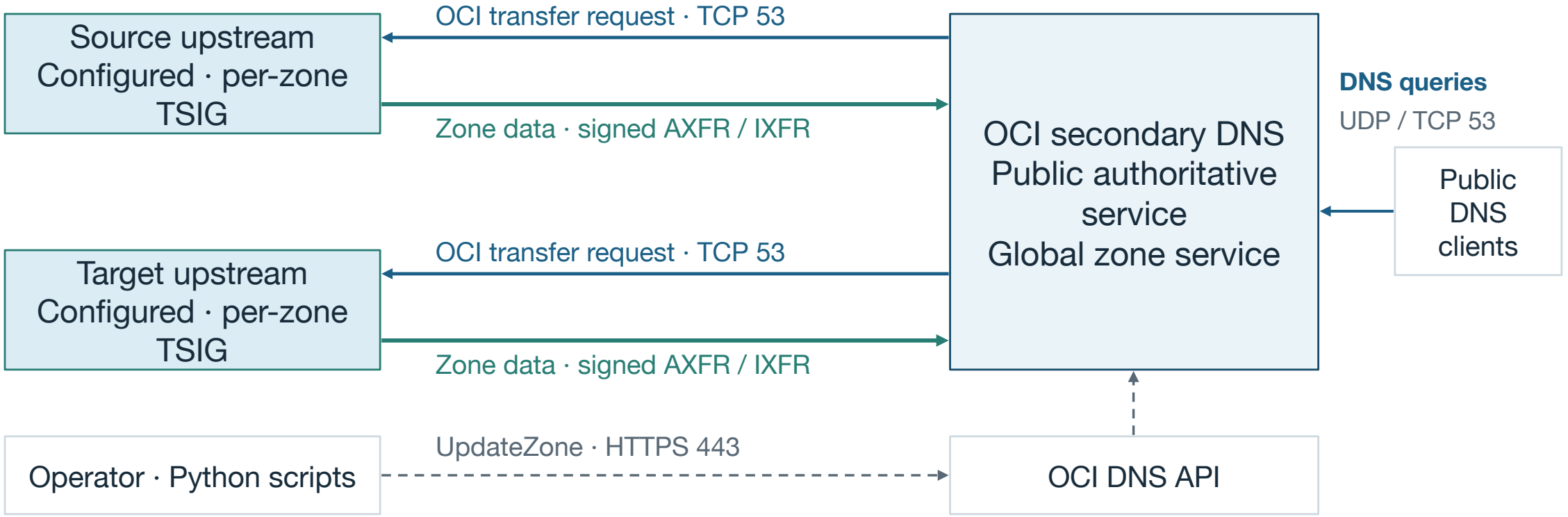
Stage	Upstream configuration	Customer gate
Baseline	Source	Target data and TSIG readiness
Phase 2	Source and target	Fresh data and target transfer checks
Phase 3	Target	Another fresh update and verification

HLD: migration flow

Click the phase links to follow the traffic paths

HLD overview | Phase 1 · source | Phase 2 · both | Phase 3 · target

Click a phase tab or Next phase to move between these PDF pages.



Per-zone TSIG on both upstreams. Customer-managed record synchronization is required.

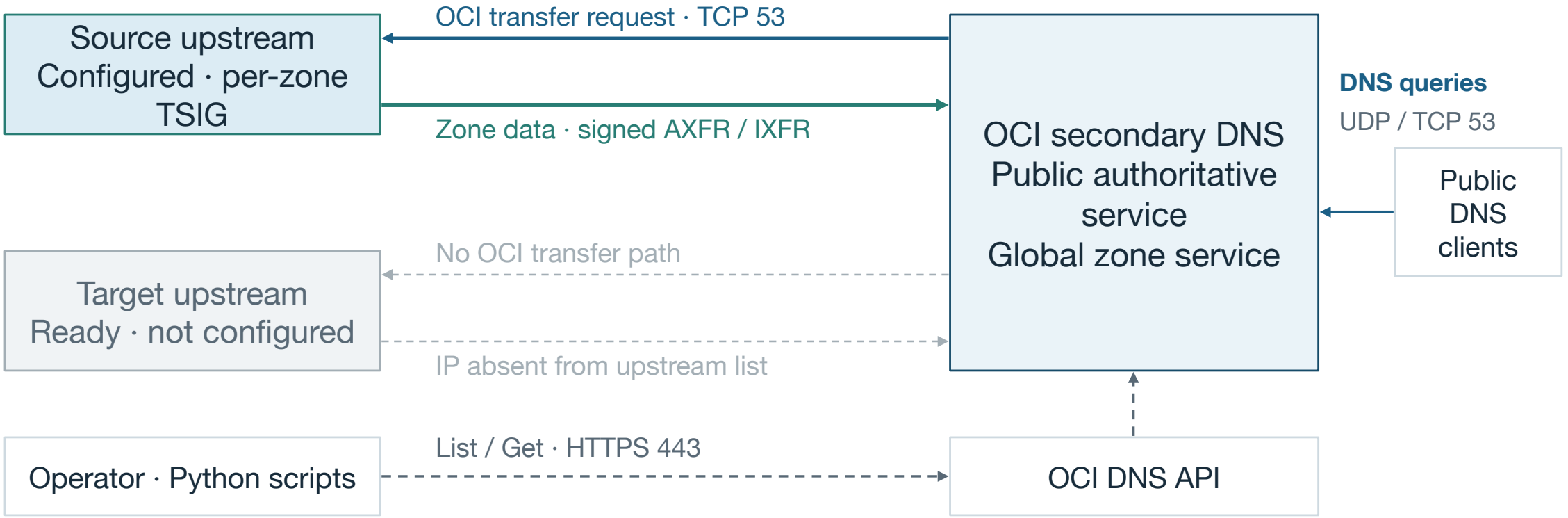
Next phase ›

HLD phase 1: source upstream only

Review the selected zone OCIDs and prepare the target

HLD overview | **Phase 1 · source** | Phase 2 · both | Phase 3 · target

Click a phase tab or Next phase to move between these PDF pages.



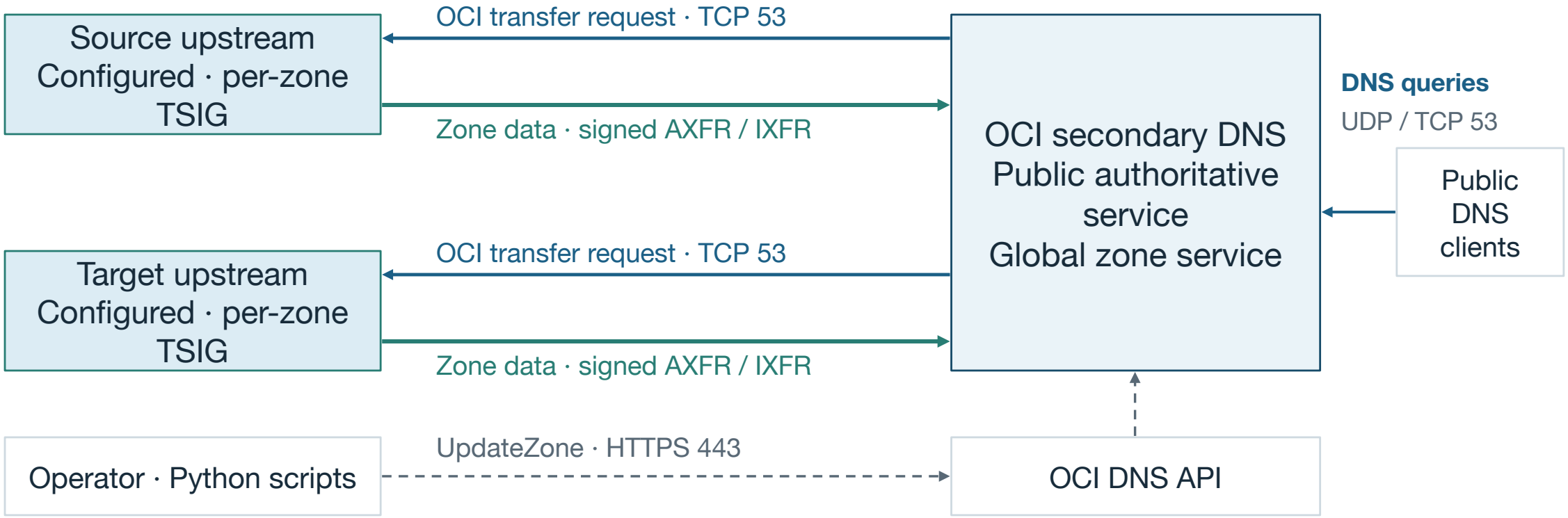
Gate: review OCID file, target records, SOA serials and per-zone TSIG before adding the target.

Next phase ›

HLD phase 2: source and target

Add the target, then verify a fresh transfer from it

Click a phase tab or Next phase to move between these PDF pages.



Gate: publish fresh data; target transfer logs prove OCI used the target during overlap.

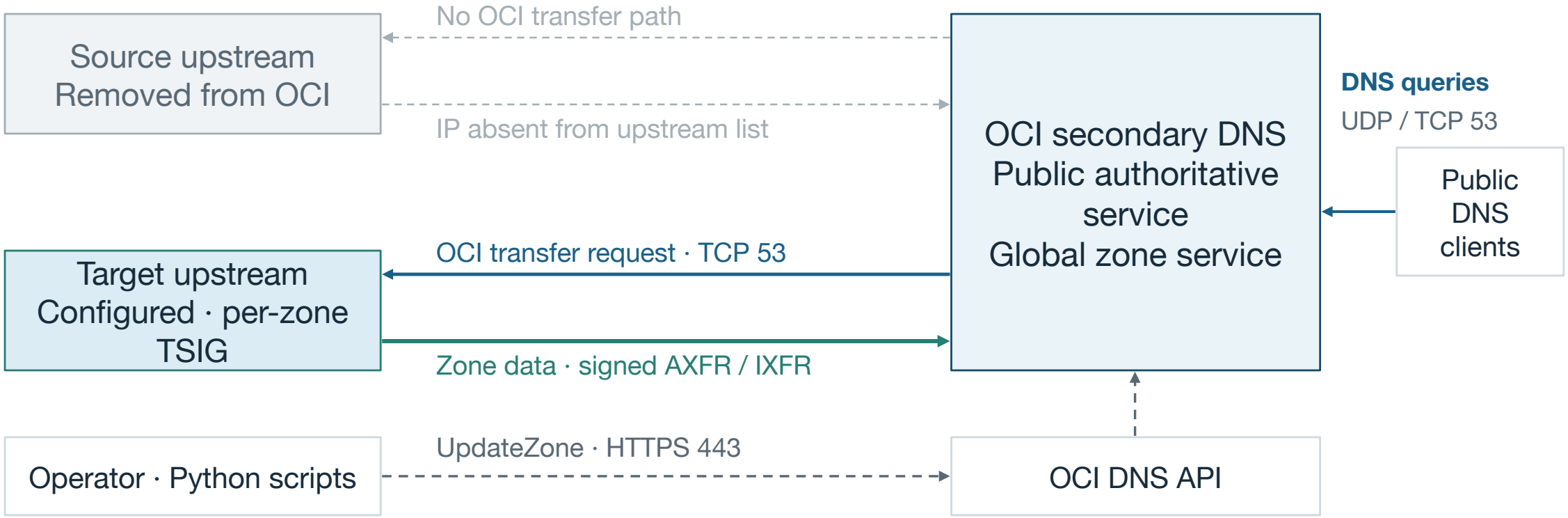
Next phase ›

HLD phase 3: target upstream only

Remove the source, then verify another fresh transfer

HLD overview | Phase 1 · source | Phase 2 · both | **Phase 3 · target**

Click a phase tab or Next phase to move between these PDF pages.



Gate: publish another fresh update; verify target-only settings and completed target transfers.

Continue ›

Customer prerequisites

Readiness before the first update

OCI scope	ACTIVE public secondary zones in the intended compartment and region.
Target readiness	Matching records, SOA serials and per-zone TSIG material.
Operator access	Python 3.11, an API-key profile and DNS zone/key permissions.
Change control	Reviewed selection, backups and a pause on concurrent updates.

Network access

Both OCI and the operator need the required DNS paths

From	To	Purpose
Operator	OCI DNS API	HTTPS 443
Operator	Target upstream	TCP 53 for signed AXFR
Operator	OCI nameservers	UDP/TCP 53 for SOA and TXT
OCI transfer servers	Upstream servers	DNS refresh and zone transfers
Upstream servers	OCI transfer servers	Signed NOTIFY on UDP 53

Authentication and TSIG

Separate credentials serve separate purposes

Credential	Purpose	Handling
API private key	OCI HTTPS authentication	Operator storage outside Git
TSIG key	Authenticated zone transfers	Matching material on both upstreams
SSH key, if used	Server administration and logs	Separate from the API key

Operator setup

A Python virtual environment keeps dependencies together

Mac and Linux

```
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install \
  -r requirements.txt
```

Windows PowerShell

```
py -3.11 -m venv .venv
$Py = ".\.venv\Scripts\python.exe"
& $Py -m pip install `
  -r requirements.txt
```

The Python SDK reads the selected OCI API-key profile.

OCI profile setup

The scripts use the selected API-key profile

oci setup config

```
[DEFAULT]
user=YOUR_USER_OCID
tenancy=YOUR_TENANCY_OCID
fingerprint=YOUR_PUBLIC_KEY_FINGERPRINT
key_file=/absolute/path/to/private_key.pem
region=YOUR_DNS_MANAGEMENT_REGION
```

Register the matching public key and verify IAM access.

Migration configuration

One non-secret JSON file supplies reusable inputs

```
{
  "compartment_id": "YOUR_COMPARTMENT_OCID",
  "profile": "DEFAULT",
  "region": "YOUR_DNS_MANAGEMENT_REGION",
  "source_ip": "192.0.2.10",
  "target_ip": "192.0.2.20",
  "zone_ids_file": "zone-ids.txt",
  "name_prefix": "example",
  "expected_count": 2,
  "output_dir": "results"
}
```

Copy examples/migration.example.json to local/migration.json.

Phase 1: zone selection

Export the selected zone OCIDs for manual review

```
python scripts/phase1_export_zones.py \
  --config local/migration.json
```

The export writes an OCID file and a separate inventory.

Selection options	Literal prefix or exact names from a text file
Read-only behavior	Missing names and count mismatches stop export

Phase 1: review gate

The reviewed file controls the later batch operations

Zone identity	Exact customer zone names and OCIDs match the intended change.
Scope and state	Every selected zone is ACTIVE, GLOBAL and SECONDARY.
Source and target	Source is signed, and target records and keys are ready.
Selection integrity	The expected count matches, with one unique OCID per line.

Phase 2: preview and apply

The target joins the source during the overlap window

Preview

```
python scripts/phase2_add_upstream.py \
  --config local/migration.json
```

Apply after review

```
python scripts/phase2_add_upstream.py \
  --config local/migration.json --apply
```

Review the planned IP/key changes before the apply command.

Phase 2: update safeguards

Safeguards for each complete upstream list

Preflight	Validate every selected zone before the first write.
Backup	Save complete original upstream lists and the selection hash.
Concurrency	Re-read settings and use a fresh ETag with If-Match.
Read-back	Wait for ACTIVE and confirm the requested upstream list.
Partial results	Stop on an error and retain results from earlier updates.

Fresh transfer verification

The DNS administrator publishes the new record value

Fresh marker	Use a unique value at _oci-transfer-check in every selected zone.
SOA serial	Increase each zone’s current serial through the DNS process.
Synchronization	Wait for the target to serve the same new records.
Refresh	Send signed NOTIFY as supported, then run verification.

Phase 2: verification command

Signed target transfer capability and fresh OCI answers

```
python scripts/phase2_verify_upstream.py \
  --config local/migration.json \
  --marker YOUR_FRESH_PHASE2_MARKER
```

Expected data result: capability-and-data-pass

Target check	Signed AXFR uses each zone’s OCI TSIG resource.
OCI check	Every assigned nameserver must match marker and SOA.

Transfer evidence

Different checks establish different facts

Check	Evidence
OCI API read-back	The intended upstream configuration is active
Operator signed AXFR	The target can serve an authenticated transfer
Fresh SOA and TXT	Target and OCI authoritative contents match
Completed target transfer log	OCI fetched that serial from the target

OCI upstream selection

Overlap requires consistent source and target data

Both addresses	Treat both configured upstreams as eligible transfer sources.
List order	Use transfer telemetry to establish the actual selected source.
NOTIFY sender	A notification requests a refresh without proving target usage.
Target-only stage	New records and target logs provide the final transfer gate.

Phase 3: preview and apply

Source removal follows the reviewed verification gate

Preview

```
python scripts/phase3_remove_upstream.py \
  --config local/migration.json
```

Apply after review

```
python scripts/phase3_remove_upstream.py \
  --config local/migration.json --apply
```

The target and unrelated upstream entries remain.

Phase 3: fresh verification

A second unique marker confirms the post-removal state

```
python scripts/phase3_verify_target.py \
  --config local/migration.json \
  --marker YOUR_FRESH_PHASE3_MARKER
```

Default requirement: source absent and target as sole upstream.

New data	Publish another marker and higher serial before this check.
Transfer proof	Add target logs for data-and-transfer-pass.

Mac terminal rehearsal

The detailed sequence is in docs/MAC_TERMINAL.md

```
cd /path/to/oci-dns-bulk-upstreams
source .venv/bin/activate
python -m unittest discover -s tests -v
nc -vz -G 5 TARGET_IP 53
```

Use a private customer configuration and review each preview.

Configuration	Compartment, region, profile, IPs and selection are customer inputs.
Connectivity	Allow the Mac's actual outbound address for signed target AXFR.

Results and recovery

The report status and proof level guide the next action

Output or status	Meaning
before.jsonl	Complete upstream lists before this invocation
results.jsonl and summary.csv	Per-zone preview, updated, unchanged or error
Verification JSON	SOA/TXT results and optional transfer evidence
Exit 1	Inspect failure and current state before retrying
Verification exit 2	Fresh data passed, target transfer proof is incomplete

Rollback process

Recovery uses current state and the relevant complete backup

Inspect	Compare current settings with results and the applicable backup.
Reconcile	Review any concurrent edits before choosing the restored list.
Restore	Submit the complete intended list using a current ETag.
Verify	Check healthy upstreams and another fresh record update.

Deployment acceptance

A controlled zone rehearsal precedes wider customer rollout

Initial rehearsal	Validate one selected zone, then a small reviewed batch.
Change gates	Require successful read-back and fresh authoritative records.
Operational review	Inspect transfer failures and zone expiry alongside server logs.
Execution evidence	Record the customer environment and acceptance outcomes.

Customer handover

Repository content supports implementation and operation

Main guide	README.md: prerequisites, inputs and phase commands
Code reference	CODE_EXPLANATION.md: functions and safeguards
Mac walkthrough	MAC_TERMINAL.md: local setup and phase rehearsal
Verification	GENERIC_VERIFICATION.md: signed transfers and evidence
Recovery	VERIFY_AND_ROLLBACK.md: partial runs and restoration

Reference documentation

Click a source name to open the official documentation

OCI secondary DNS	Transfer roles, TSIG and secondary operation
Upstream updates	Per-zone Console, CLI and API update workflow
ExternalMaster model	IPv4/IPv6 address, port 53 and TSIG reference
Access and monitoring	API signing keys, DNS policies and transfer metrics